

Task1 & Task 2:

```
#include<iostream>
using namespace std;
int main(){
    string ex = "a = b * c + d / e - f";
    cout<<"Expression : "<<ex;
    cout<<"\nTemp1 = b * c"<<endl;
    cout<<"Temp 2 = d / e"<<endl;
    cout<<"Temp3 = Temp1 + Temp2"<<endl;
    cout<<"a = Temp3 - f"<<endl;

    cout<<"Optimized Version : "<<endl;
    cout<<"Temp1 = b * c"<<endl;
    cout<<"Temp2 = Temp1 + d/e"<<endl;
    cout<<"a = Temp2 - f"<<endl;
}
```

Task3:

```
#include <iostream>
#include <string>
using namespace std;
int main() {
    string obj1[2] = {"add", "subtract"};
    string obj2[2] = {"multiply", "divide"};

    cout<<"Object File 1:"<<endl;
    for(int i = 0; i < 2; i++){
        cout << " - " << obj1[i]<<endl;
    }
    cout << "Object File 2 :"<<endl;
    for(int i = 0; i < 2; i++){
        cout << " - " << obj2[i]<<endl;
    }

    cout<<"Executable :"<<endl;
    for(int i = 0; i < 2; i++){
        cout << " - " << obj1[i] << "()"<<endl;
    }
    for (int i = 0; i < 2; i++) {
        cout << " - " << obj2[i] << "()\n";
    }

    return 0;
}
```

Task 4:

```
#include<iostream>
using namespace std;
int main(){
    string s;cin>>s;
    if(s.length()>3){
        cout<<"Invalid";return 0;
    }
    else if(s[0]=='+'&& s[1] =='+'){
        cout<<"valid"<<endl;
        return 0;
    }
    else if((s[1]=='+' && s[2] == '+')){
        cout<<"valid"<<endl;
        return 0;
    }
    else if(s[0]=='-'&& s[1] =='-'){
        cout<<"valid"<<endl;
        return 0;
    }
    else if((s[1]=='-' && s[2] == '-')){
        cout<<"valid"<<endl;
        return 0;
    }

    else{
        cout<<"Invalid"<<endl;
    }
}
```